25900237 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCBIF201 - APPLIED MATHEMATICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Define a singleton set with an example. [K] / [CO1]
2. Which law is applied in: $(Y \vee F) \equiv Y$? [U] / [CO3]
3. Define adjacency matrix and explain its structure with an example [K] / [CO5]
4. What is a p-value? [K] / [CO8]
5. State whether tautology, contradiction, and contingency are mutually exclusive. Explain shortly [U] / [CO3]
6. Compare distributive law in sets with distributive law in arithmetic. [U] / [CO2]
7. What is logical equivalence [U] / [CO3]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

8. Explain why a connected graph with *n* vertices and *(n–1)* edges must be a tree [U] / [CO6]

9. Differentiate between bar chart, histogram, and pie chart with suitable diagrams. [U] / [CO8]

10. Consider a graph with 5 vertices and 6 edges. Draw a possible undirected graph and label the vertices [U] / [CO5]

11. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, compute A×I and I×A where I is the identity matrix. [U] / [CO2]

12. State the commutative laws of logic with suitable examples [U] / [CO4]

13. Write short notes on Hamiltonian cycle and give an example graph which has it [K] / [CO6]

14. A coin is tossed 3 times. Find the probability distribution of the number of heads. [A] / [CO8]

15. Construct the truth table for $(p \wedge q) \wedge \neg q$ [A] / [CO3, CO4]

[5x6 = 30]

Part C

Long Essay Questions

Answer any **3** questions

Each question carries **10** marks

16. Solve by matrix method: [K] / [CO7]

x+y=5,2x−y=1,3x+y=7

17. Define a conditional statement with suitable examples. Explain its truth table clearly. [U] / [CO4]

18. Define connected graphs. Explain the difference between connected and disconnected graphs with diagrams. [K] / [CO5]

19. Define and distinguish between Mean, Median, and Mode. Explain with suitable examples. [K] / [CO8]

20. What is meant by logic in mathematics? Why is it considered important in solving mathematical problems? [U] / [CO3]

[10x3 = 30]